STATISTICS 290

Statistical Consulting

Please introduce yourself: can follow the following questions… (around 15 only)

- Name
- Affiliation (current)
- What are your expectations from this course?

What is Statistical Consulting?

- **A**ny interpersonal interaction
  that has the goal of increasing the utility of information
  disseminated within and among disciplines
  that uses STATISTICS.

- **The Practice of Statistics**

Course Objectives

- mechanics and protocols of statistical consulting
  - communicate to clients=> understand the research problem
  - convert the layman's problem into a statistical problem

    => well-defined statistical problem
  - statistical computing
  - present results to answer the questions

Reminders

- Classes start at 6:00 PM
- Cellular phones should be TURNED OFF before the start of the sessions.

Course Plan

- Introduction
  - Statistical Consulting Process
  - The Practice of Statistics
  - Statistical Rules of Thumb
  - Tukey, EDA, The Future of Data Analysis
  - Marquardt and Statistical Consulting
  - Philosophy of Statistics

Course Plan

- Introduction
  - Other Topics, e.g.,
    - Forecasting
    - Data Mining and Business Intelligence
    - Monte Carlo Methods and Simulation
    - Statistical Machine Learning

Course Plan

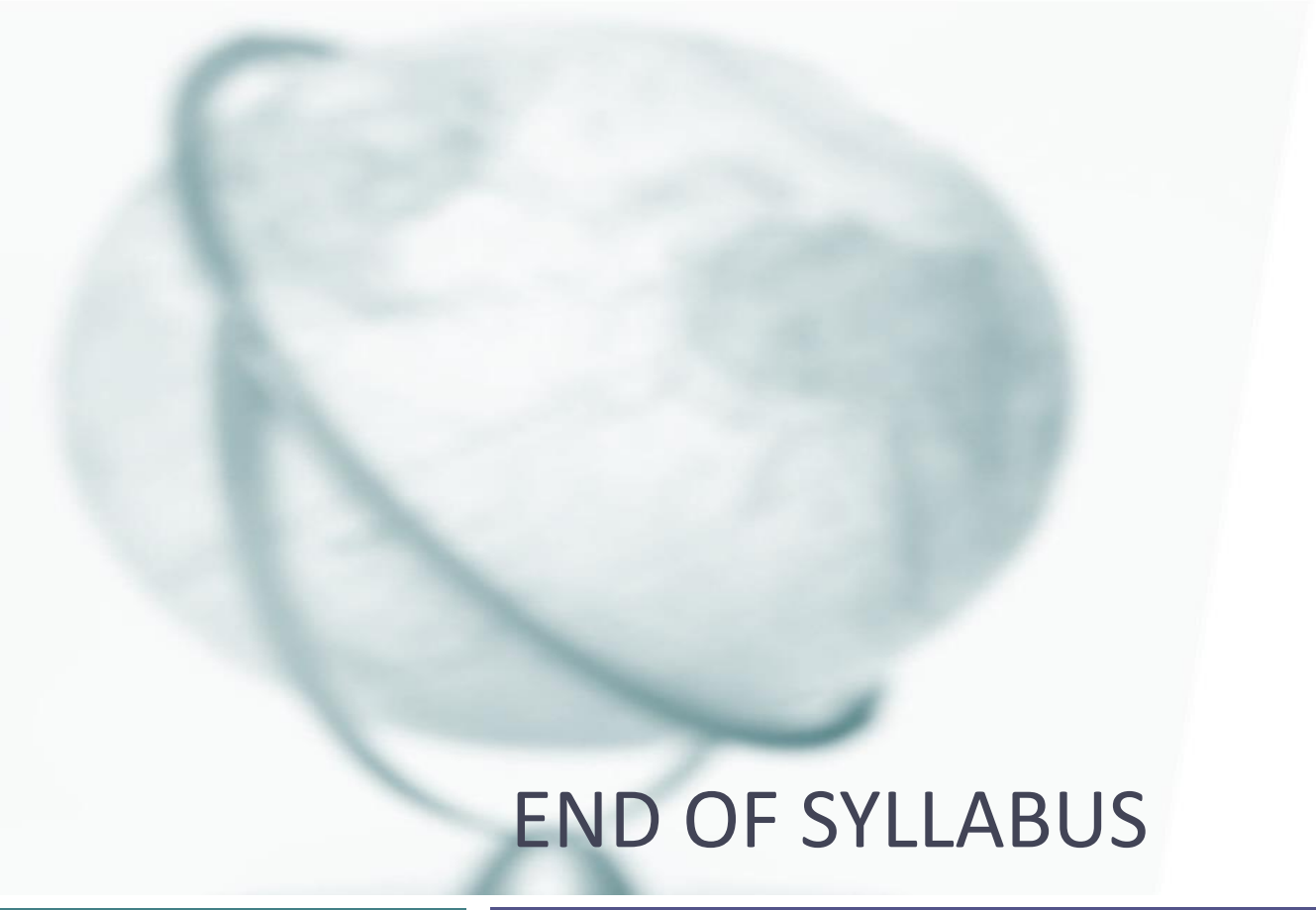
- Cases
  - Data Mining, Analytics
  - Consumer Behavior Modeling
  - Clinical Trials
  - Customer Satisfaction Studies
  - WTP Survey
  - Impact Assessment

Course Plan

- Recent Advances in Statistics
  - Nonparametric Modeling
  - High Dimensional Data
  - Clustered Data
  - Hierarchical Modeling

Requirements

- To get a grade of **P** (passed), one should:
  - Attend in at least 75% of class meetings.
  - Participate in class discussions.
  - **Answer questions.**
  - **ASK questions.**



END OF SYLLABUS